

Virtaul Lab Demo

1.Depth First Search

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ai1-iith.vlabs.ac.in/exp/ai-depth-first-search/simulation.html

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Depth First Search

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Tasks Depth First Search

Instructions

Controls

Start Vertex: 5
Start DFS Next
☒ AutoPlay
Toggle Forces Clear
Clear Visited
Note: Overwrites Graph
Tree Controls:
Tree Depth: 3

Information

Frontier: []
Visiting: undefined
To Visit: undefined

```
graph TD; 0((0)) --- 1((1)); 0 --- 2((2)); 1 --- 3((3)); 1 --- 4((4)); 2 --- 5((5)); 2 --- 6((6));
```

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2. Greedy Best First search

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Greedy Best First Search

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Tasks Best First Search

Instructions

Controls

Start Vertex: 4
Goal Vertex: 6
Start BEFS Next
☒ AutoPlay
Toggle Forces Clear
Clear Visited
Note: Overwrites Graph
Tree Controls:
Tree Depth: 3

Information

Goal Vector: 6
Frontier: [5,3]
Estimated Cost(E): [248, 743]
Visiting: undefined
To Visit: 5

Observation

Found Goal!

```
graph TD; 0((0)) -- 141 --> 1((1)); 0 -- 254 --> 2((2)); 1 -- 158 --> 3((3)); 1 -- 272 --> 4((4)); 2 -- 234 --> 5((5)); 2 -- 117 --> 6((6));
```

Gold
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[illegible]

Iterative Deepening Depth First Search

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Tasks

Iterative Deepening Depth First Search

Instructions

Controls

Start Vertex:

End Vertex:

Start IDDFS

☒ AutoPlay

Note: Overrides Graph Tree Controls:
| Tree Depth:

```
graph TD
    0((0)) --- 1((1))
    0 --- 2((2))
    1 --- 3((3))
    1 --- 4((4))
    2 --- 5((5))
    2 --- 6((6))
```

Information

- Frontier: []
- Visiting:
- To Visit:
- Depth Limit:
- Goal found

5. A* Search

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ai2-iith.vlabs.ac.in/exp/a-star-search/simulation.html

Virtual Labs A* Search ☆☆☆☆☆ Rate Me Report a Bug

Tasks Graph 1 Graph 2 Graph 3

Instructions

Controls

Start Vertex: 4
End Vertex: 1
Start A* Next
AutoPlay
Clear Visited

Heuristics Table

Node 0: 0
Node 1: 69
Node 2: 47
Node 3: 91
Node 4: 109

Information

Frontier: []
Visiting: 0 (parent = 2)
To Visit: none
Current Path: 4 -> 3 -> 2 -> 0

Path History

Step 1: 4 (g=0, h=109, f=109, edges=)
Step 2: 4 -> 1 (g=99, h=69, f=168, edges=99)
Step 3: 4 -> 3 (g=80, h=91, f=171, edges=80)

Tomorrow's low To tie record

Search

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6. Hill Climbing Search

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Virtual Labs Hill Climbing ☆☆☆☆☆ Rate Me Report a Bug

Controls

Grid Size: 8x8 (1)
Sideways Move
Start Next
AutoPlay
Reset

Explanation

Information

Iteration: 5
Best Heuristic Value: 2

History

8 Queens Problem

Iteration: 0 BHV: 11
Iteration: 1 BHV: 7
Iteration: 2 BHV: 4

Local Minima

The algorithm is stuck in a local minima.

Grid:

(0,0)	3	(0,1)	3	(0,2)	3	(0,3)	2	(0,4)	3	(0,5)	4	(0,6)	4	(0,7)	4
(1,0)	4	(1,1)	3	(1,2)	4	(1,3)	2	(1,4)	5	(1,5)	4	(1,6)	4	(1,7)	3
(2,0)	5	(2,1)	4	(2,2)	2	(2,3)	4	(2,4)	3	(2,5)	6	(2,6)	4	(2,7)	4
(3,0)	5	(3,1)	3	(3,2)	4	(3,3)	3	(3,4)	4	(3,5)	3	(3,6)	6	(3,7)	5
(4,0)	4	(4,1)	4	(4,2)	4	(4,3)	4	(4,4)	2	(4,5)	4	(4,6)	4	(4,7)	5
(5,0)	5	(5,1)	4	(5,2)	4	(5,3)	4	(5,4)	5	(5,5)	5	(5,6)	5	(5,7)	4
(6,0)	3	(6,1)	2	(6,2)	2	(6,3)	3	(6,4)	3	(6,5)	3	(6,6)	2	(6,7)	4
(7,0)	4	(7,1)	2	(7,2)	3	(7,3)	4	(7,4)	3	(7,5)	4	(7,6)	3	(7,7)	3

Tomorrow's low To tie record

Search

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Feedback

Inhouse Feedback

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Your current semester: 5th Semester

+

S.No	Discipline	Lab Name	Experiment Name	Delete
1	Computer Science & Engineering	Artificial Intelligence I	AI Depth First Search	
2	Computer Science & Engineering	Artificial Intelligence I	Greedy Best First Search	
3	Computer Science & Engineering	Artificial Intelligence I	Minimax Search	
4	Computer Science & Engineering	Artificial Intelligence II	Iterative Deepening Depth First Search	
5	Computer Science & Engineering	Artificial Intelligence II	A* Search	
6	Computer Science & Engineering	Artificial Intelligence II	Hill Climbing Search	

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Please indicate your agreement with the following statements:

a. The degree to which the actual lab environment is simulated ☒ Excellent ☐ Very Good ☐ Good ☐ Fair ☐ Poor

b. The manuals were to be found helpful ☒ Excellent ☐ Very Good ☐ Good ☐ Fair ☐ Poor

c. The results of experiment were easily interpretable ☒ Excellent ☐ Very Good ☐ Good ☐ Fair ☐ Poor

Please tell your agreement with the following statements:

d. What was your experience while performing the experiment in virtual lab? ☒ Excellent ☐ Very Good ☐ Good ☐ Fair ☐ Poor

e. Do you think performing experiments through Virtual Labs gives scope for more innovative and creative research work ☒ Yes ☐ Maybe ☐ To some extent ☐ Don't know ☐ No

f. Did you go through the manual / step by step method before performing the live experiments ☒ Yes ☐ No

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